



United International University

School of Science and Engineering

Final Examination; Year 2025; Trimester: Summer

Course: BIO 3105/3107; Title: Biology for Engineers/Biology;

Sec: All

Full Marks: 40; Time: 2 hours

There are Five Questions, **1, 2, and 3 are mandatory** to answer, and answer **4 or 5** (anyone).

Any examinee found adopting unfair means will be expelled from the trimester/program as per UIU disciplinary rules

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| 1. | a. Any biological proteins can be produced in bacteria? Name the method and briefly describe the workflow. | 4 | CO4 |
| | b. Immune system is smart and remember the pathogens once they encounter. Discuss how? | 2 | CO1 |
| | c. “A healthy balanced diet can make you happy”– Justify the statement. | 2 | CO3 |
| | d. Define “Antibody” and “Antigen”? Mention different types of antigens. | 2 | CO2 |
| 2. | a. From gene cloning to virus detection, DNA amplification is essential. Illustrate the process of how you can do that. | 4 | CO3 |
| | b. Define “Food Web” and “Food Chain”? Comment on the statement. | 2 | CO2 |
| | c. What are the different types of vaccine? Justify why mRNA vaccine are advantageous than other types of vaccine. | 4 | CO4 |
| 3. | a. Define and differentiate “innate” and “adaptive” immunity? | 4 | CO4 |
| | b. How antibody therapy can be useful to treat severe infections. Justify. | 2 | CO2 |
| | c. What should be the change in energy flow for a typical ecosystem where there are 3 levels of consumers? Show with the help of energy-time graph. | 4 | CO1 |
| 4. | a. Biotechnological methods can be employed in forensic science and criminal identification. Describe how? | 5 | CO4 |
| | b. Mention the relationship between food and stress management. | 5 | CO3 |
| 5. | a. Are proteins important in our diet plan? In a balanced diet how much protein should be present and why? | 5 | CO2 |
| | b. If a child is born with a genetic disorder, describe how you can treat him/her with modern Biotechnological tools and applications. | 5 | CO1 |

CO1: Describe different biological quantities.

CO2: Apply the knowledge of biological systems in a real-life problem.

CO3: Design several biological systems with constraints.

CO4: Explain several procedures for solving biological systems within constraints.